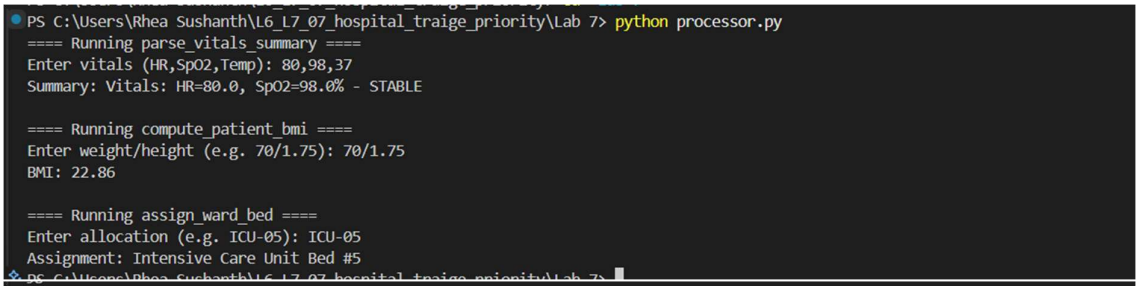
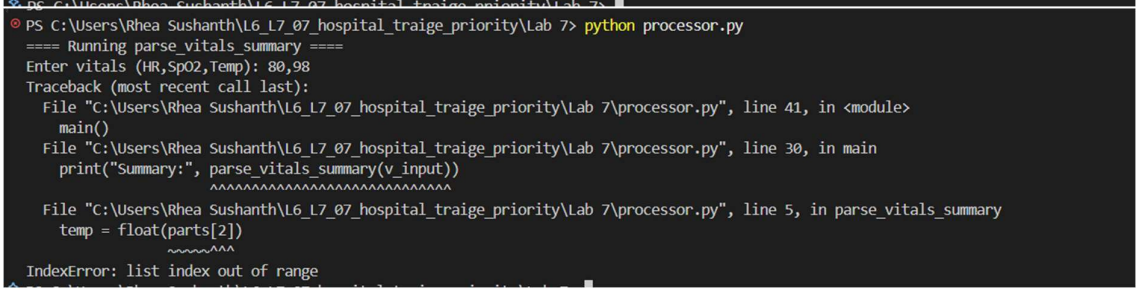
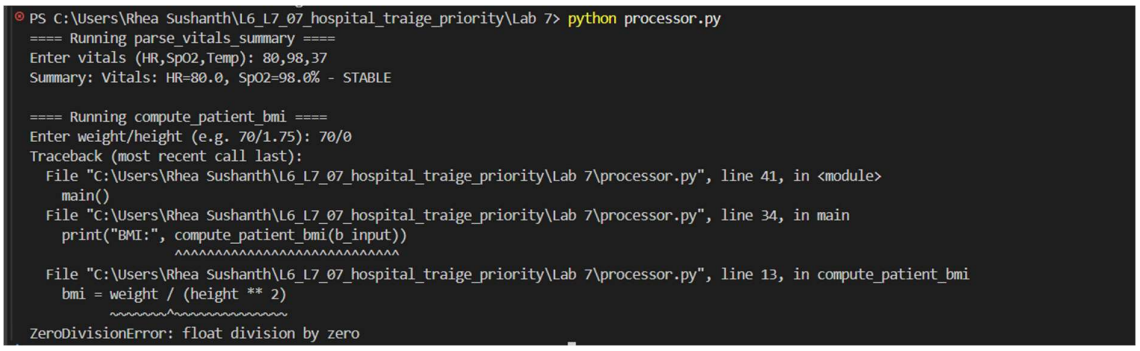
LAB7

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SRN: PES1UG24CS371







Updated test_processor.py

```
from hypothesis import given, strategies as st
from processor import parse_vitals_summary, compute_patient_bmi, assign_ward_bed

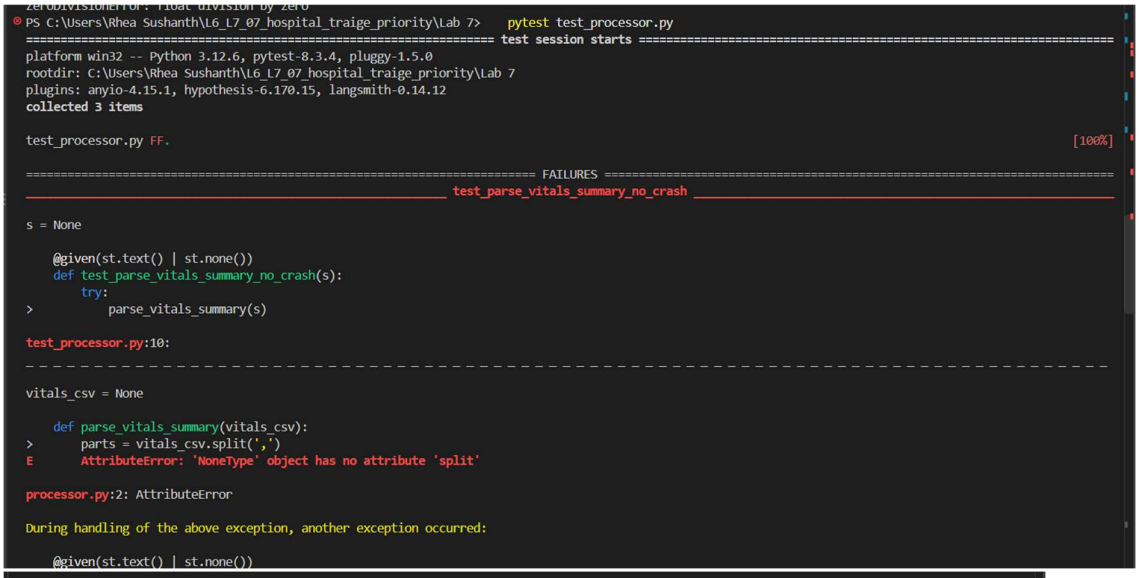
bmi_inputs = st.builds(lambda w, h: f"{w}/{h}", st.floats(), st.floats())
ward_inputs = st.builds(lambda w, n: f"{w}-{n}", st.text(), st.text())

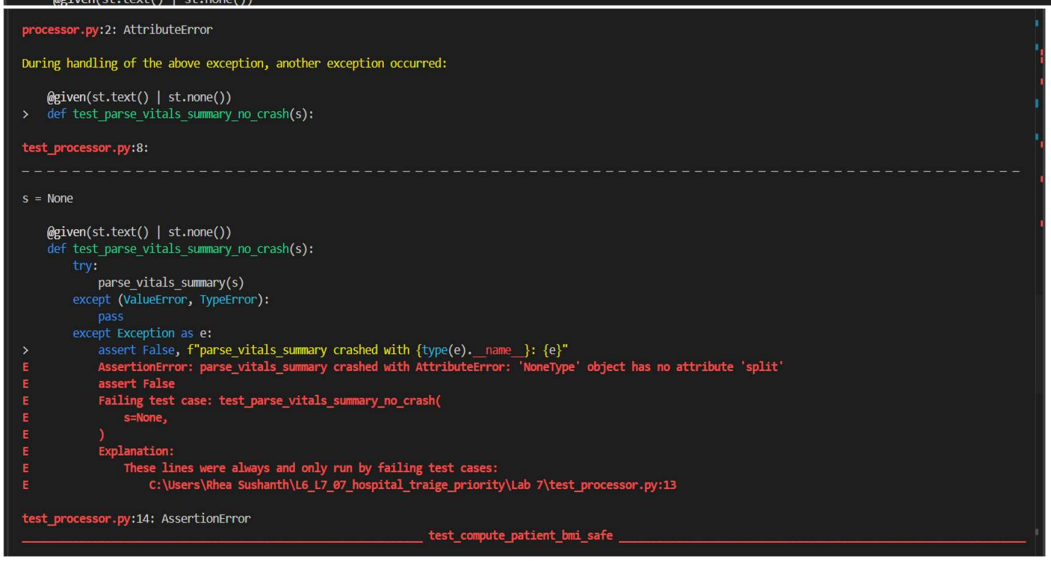
@given(st.text() | st.none())
def test_parse_vitals_summary_no_crash(s):
    try:
        parse_vitals_summary(s)
    except (ValueError, TypeError):
        pass
    except Exception as e:
        assert False, f"parse_vitals_summary crashed with {type(e).__name__}: {e}"

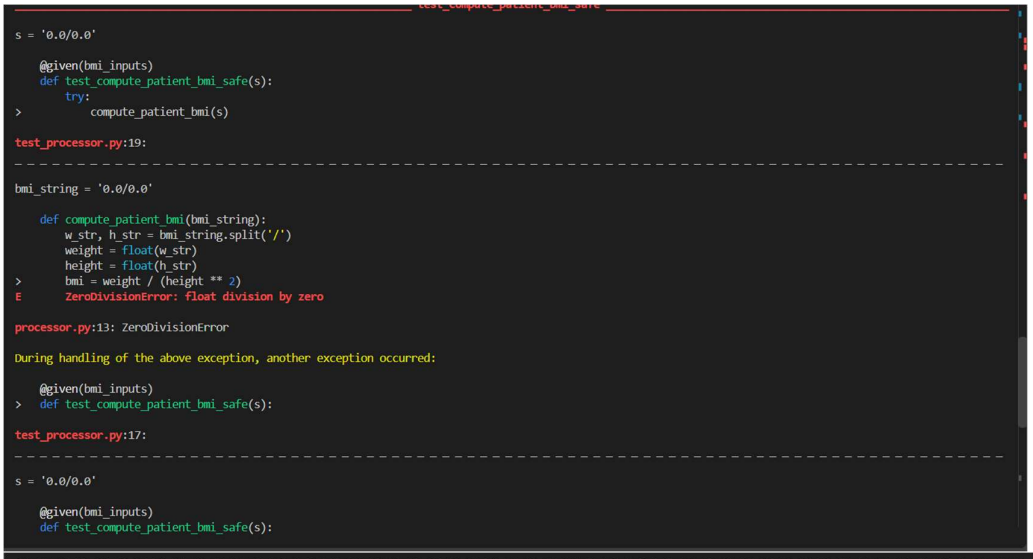
@given(bmi_inputs)
```

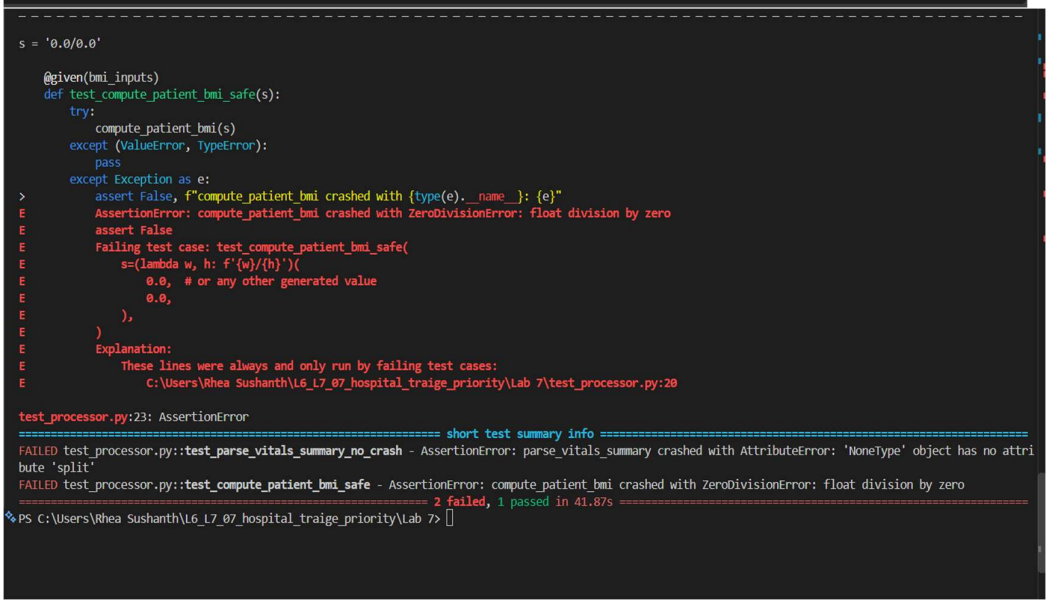
```
def test_compute_patient_bmi_safe(s):
    try:
        compute_patient_bmi(s)
    except (ValueError, TypeError):
        pass
    except Exception as e:
        assert False, f"compute_patient_bmi crashed with {type(e).__name__}: {e}"

@given(ward_inputs)
def test_assign_ward_bed_safe(s):
    try:
        assign_ward_bed(s)
    except (ValueError, TypeError):
        pass
    except Exception as e:
        assert False, f"assign_ward_bed crashed with {type(e).__name__}: {e}"
```









Processor.py

```
import math

def parse_vitals_summary(vitals_csv):
    if not isinstance(vitals_csv, str):
        raise ValueError("Vitals input must be a string")

    parts = vitals_csv.split(',')
    if len(parts) != 3:
        raise ValueError("Vitals must have exactly 3 values: HR,SpO2,Temp")

    try:
        hr = float(parts[0])
        spo2 = float(parts[1])
        temp = float(parts[2])
```

```
    except ValueError:
      raise ValueError("Vitals values must be numbers")

    if not all(math.isfinite(v) for v in (hr, spo2, temp)):
      raise ValueError("Vitals values must be finite numbers")

    critical = hr > 120 or spo2 < 90 or temp > 39.0
    return f"Vitals: HR={hr}, SpO2={spo2}% - {'CRITICAL' if critical else 'STABLE'}"

  def compute_patient_bmi(bmi_string):
    if not isinstance(bmi_string, str):
      raise ValueError("BMI input must be a string")

    parts = bmi_string.split('/')
    if len(parts) != 2:
      raise ValueError("BMI input must be in the form weight/height")

    try:
      weight = float(parts[0])
      height = float(parts[1])
    except ValueError:
      raise ValueError("Weight and height must be numbers")

    if not (math.isfinite(weight) and math.isfinite(height)):
      raise ValueError("Weight and height must be finite numbers")
    if weight <= 0 or height <= 0:
      raise ValueError("Weight and height must be greater than zero")

    try:
      bmi = weight / (height ** 2)
    except (ZeroDivisionError, OverflowError):
      raise ValueError("Height is outside the valid range")

    if not math.isfinite(bmi):
      raise ValueError("BMI result is not a finite number")

    return round(bmi, 2)

  def assign_ward_bed(allocation_code):
    if not isinstance(allocation_code, str):
      raise ValueError("Allocation code must be a string")

    parts = allocation_code.strip().split('-')
    if len(parts) != 2:
      raise ValueError("Allocation code must be in the form WARD-BED")
    ward, bed_str = parts

    ward_names = {
```

```
        "ER": "Emergency Room",
        "ICU": "Intensive Care Unit",
        "GEN": "General Ward",
        "PED": "Pediatrics"
    }

    ward_key = ward.strip().upper()
    if ward_key not in ward_names:
        raise ValueError(f"Unknown ward: {ward}")

    try:
        bed_num = int(bed_str)
    except ValueError:
        raise ValueError("Bed number must be a whole number")

    if bed_num <= 0:
        raise ValueError("Bed number must be positive")

    return f"{ward_names[ward_key]} Bed #{bed_num}"

def main():
    print("==== Running parse_vitals_summary ====")
    v_input = input("Enter vitals (HR,SpO2,Temp): ")
    print("Summary:", parse_vitals_summary(v_input))

    print("\n==== Running compute_patient_bmi ====")
    b_input = input("Enter weight/height (e.g. 70/1.75): ")
    print("BMI:", compute_patient_bmi(b_input))

    print("\n==== Running assign_ward_bed ====")
    w_input = input("Enter allocation (e.g. ICU-05): ")
    print("Assignment:", assign_ward_bed(w_input))

if __name__ == "__main__":
    main()
```

```
PS C:\Users\Rhea Sushanth\L6_L7_07_hospital_traige_priority\Lab 7> python processor.py
==== Running parse_vitals_summary ====
Enter vitals (HR,SpO2,Temp): 80,98
Traceback (most recent call last):
  File "C:\Users\Rhea Sushanth\L6_L7_07_hospital_traige_priority\Lab 7\processor.py", line 97, in <module>
    main()
  File "C:\Users\Rhea Sushanth\L6_L7_07_hospital_traige_priority\Lab 7\processor.py", line 86, in main
    print("Summary:", parse_vitals_summary(v_input))
                      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Rhea Sushanth\L6_L7_07_hospital_traige_priority\Lab 7\processor.py", line 9, in parse_vitals_summary
    raise ValueError("Vitals must have exactly 3 values: HR,SpO2,Temp")
ValueError: Vitals must have exactly 3 values: HR,SpO2,Temp
```

```
PS C:\Users\Rhea Sushanth\L6_L7_07_hospital_traige_priority\Lab 7> python processor.py
==== Running parse_vitals_summary ====
Enter vitals (HR,SpO2,Temp): 80,98,99
Summary: Vitals: HR=80.0, SpO2=98.0% - CRITICAL

==== Running compute_patient_bmi ====
Enter weight/height (e.g. 70/1.75): 70/0
Traceback (most recent call last):
  File "C:\Users\Rhea Sushanth\L6_L7_07_hospital_traige_priority\Lab 7\processor.py", line 97, in <module>
    main()
  File "C:\Users\Rhea Sushanth\L6_L7_07_hospital_traige_priority\Lab 7\processor.py", line 90, in main
    print("BMI:", compute_patient_bmi(b_input))
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Rhea Sushanth\L6_L7_07_hospital_traige_priority\Lab 7\processor.py", line 41, in compute_patient_bmi
    raise ValueError("Weight and height must be greater than zero")
ValueError: Weight and height must be greater than zero
```

```
PS C:\Users\Rhea Sushanth\L6_L7_07_hospital_traige_priority\Lab 7> pytest test_processor.py
============================= test session starts ==============================
platform win32 -- Python 3.12.6, pytest-8.3.4, pluggy-1.5.0
rootdir: C:\Users\Rhea Sushanth\L6_L7_07_hospital_traige_priority\Lab 7
plugins: anyio-4.15.1, hypothesis-6.170.15, langsmith-0.14.12
collected 3 items

test_processor.py ...                                                    [100%]

============================== 3 passed in 5.00s ===============================
PS C:\Users\Rhea Sushanth\L6_L7_07_hospital_traige_priority\Lab 7>
```